Highlights

Sparse classifier chains for robust multi-label classification

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- We improve the interpretability and robustness of classifier chains (CCs) by allowing sparse structures.
- We propose a heuristic algorithm for finding sparse CCs targeting any given evaluation metric.
- We demonstrate the advantages of our proposal via an extensive empirical study.

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ABSTRACT

Classifier chains and their ensembles have been shown to be accurate in multi-label classification (MLC). This paper is an attempt to improve the interpretability and robustness of classifier chains by allowing sparse structures, instead of the dense structures enforced by the standard classifier chains. Technically, we propose a heuristic algorithm for finding sparse classifier chains targeting any given evaluation metric. We then conduct an extensive empirical study to demonstrate the potential advantages of our proposal in the standard MLC setting and MLC with partial abstention, where the classifiers are allowed to abstain from making precise predictions on uncertain labels.

1. Introduction

Multi-label classification (MLC), where each instance can be associated with none, one, or multiple labels simultaneously, has been receiving increasing attention (García-Pedrajas, Cuevas-Muñoz, Cerruela-García and de Haro-García, 2024; Madjarov, Kocev, Gjorgjevikj and Džeroski, 2012; Nguyen, Hoang, Hoang and Huynh, 2026b; Nguyen, Hoang, Destercke, de Campos and Huynh, 2026a; Tsoumakas, Katakis and Vlahavas, 2010; Zhang and Zhou, 2013). One of the main challenges is to deal with multiple evaluation metrics, which reflect different ways to assess the multi-label classifiers (García-Pedrajas et al., 2024; Nguyen and Hüllermeier, 2021; Zhang and Zhou, 2013). A common practice in constructing MLC methods is to follow the statistical learning theory (SLT) (Vapnik, 1999): Once the evaluation metric is specified, the empirical risk minimization principle is adopted to find an optimal classifier that optimizes the empirical risk functional constructed based on the training set.

A specific family of such methods is the so-called discriminative chain rule-based classifiers, which essentially seek good estimates of the joint conditional probability distributions over the label variables by optimizing (surrogates of) 0/1 loss (Gerych, Hartvigsen, Buquicchio, Agu and Rundensteiner, 2021; Gonçalves, Plastino and Freitas, 2015; Liu and Tsang, 2015; Nguyen, Yang and De Campos, 2023; Nguyen et al., 2026b). The optimization of multiple instance-based MLC losses is then done at the prediction phase by predicting Bayes-optimal predictions (BOPs) following the Bayesian inference principle (Dembczyński, Waegeman, Cheng and Hüllermeier, 2012; Nguyen et al., 2026b; Waegeman, Dembczyński, Jachnik, Cheng and Hüllermeier, 2014). While being theoretically sound and interpretable (thanks to the compact representation of the dependencies encoded by the directed acyclic graphs (DAGs) over the label variables), this family often comes with computationally expensive prediction phases (Nguyen et al., 2023, 2026b). Classifier chains (Read, Pfahringer, Holmes and Frank, 2011, 2021) are among the most popular multi-label classifiers within this family, where the dependencies are encoded by fully connected DAGs.

A common practice to mitigate the computational burden is to proceed with the label independence assumption (Dembczyński et al., 2012; Nguyen and Hüllermeier, 2021), which results in the use of the so-called binary relevance (BR) classifiers in the training phase and allows polynomial prediction algorithms for BOP computation of various MLC evaluation metrics (Nguyen and Hüllermeier, 2021). However, this clearly comes with the cost of interpretability, especially when one wishes to see which label dependencies may benefit the optimization of the evaluation metrics.

There is another family of multi-label classifiers targeting the given specific population-based MLC evaluation metric (Pillai, Fumera and Roli, 2017; Zhang, Li, Liu and Geng, 2018). Notable attempts in this family are constructed based on BR classifiers and mainly differ in the thresholds used to make the final predictions (Pillai et al., 2017; Zhang et al., 2018). Often, the thresholds are defined at the population level, which then results in low prediction costs. While

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being scalable and rather accurate, they again come with the cost of interpretability, especially when one wishes to see which label dependencies may benefit the optimization of the evaluation metrics.

This paper is an attempt to combine the advantages of the two mentioned families of multi-label classifiers. More precisely, we propose to learn sparse classifier chains targeting the given specific population-based MLC evaluation metrics. This allows low prediction costs and, at the same time, suggests information on dependencies which may benefit the optimization of the evaluation MLC metric. Clearly, the problem of finding the most sparse classifier chain that optimizes a given specific population-based MLC evaluation metric is challenging due to the exponential number of possible DAGs over the label variables. Therefore, we propose a greedy approach to find approximate solutions. Through an extensive empirical study, we demonstrate the potential advantages of our proposal in the standard MLC setting and MLC with partial abstention, where the classifiers are allowed to abstain from making precise predictions on uncertain labels (Nguyen and Hüllermeier, 2021; Nguyen et al., 2026a).

The remainder of this paper is organized as follows. Section 2 contains an overview of classifier chains. In Section 3, we present Sparse Classifier Chains alongside the training and prediction procedures of our proposed method. To compare our approach with other multi-label classification methods, a series of experiments was conducted in Section 4.

2. Classifier chains

2.1. Generalities

Let $\mathcal{X} \subseteq \mathbb{R}^d$ denote the instance space and let $\mathcal{Y} = \{0,1\}^L$ be the output space associated with a set of labels $\mathcal{L} = \{1, \dots, L\}$. A multi-label example is a pair $(\boldsymbol{x}, \boldsymbol{y}) \in \mathcal{X} \times \mathcal{Y}$, where $y_\ell = 1$ indicates the relevance of label ℓ and $y_\ell = 0$ its irrelevance. Given a training sample $\mathcal{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^n$ drawn independently from an unknown distribution $P(\boldsymbol{X}, \boldsymbol{Y})$, the conventional MLC task is to learn a predictor $h : \mathcal{X} \to \mathcal{Y}$. Its output $\hat{\boldsymbol{y}} = h(\boldsymbol{x})$ is evaluated by an MLC metric, which may aggregate prediction outcomes across labels, instances, or both (Dembczyński et al., 2012; Zhang and Zhou, 2013). In the partial-abstention setting, the prediction space is extended to $\{0, 1, \perp\}^L$, where $\perp$ denotes that no precise prediction is issued for the corresponding label (Nguyen and Hüllermeier, 2021).

Classifier chains (CCs) transform the MLC problem into L binary classification problems and connect them according to a label order $\boldsymbol{\pi} = (\pi_1, \dots, \pi_L)$, which is a permutation of $\mathcal{L}$ (Read et al., 2011, 2021). By the probability chain rule, the conditional distribution of the label vector can be factorized for any such order as

$$P(\boldsymbol{y} \mid \boldsymbol{x}) = \prod_{j=1}^{L} P\left(y_{\pi_j} \mid \boldsymbol{x}, y_{\pi_1}, \dots, y_{\pi_{j-1}}\right). \tag{1}$$

A probabilistic CC approximates each factor in eq. (1) by a binary conditional model q_{π_j}. Thus, the classifier for label π_j uses the original features together with the labels preceding it in $\boldsymbol{\pi}$, whereas the first classifier only uses $\boldsymbol{x}$.

Equivalently, a standard CC can be represented by a directed acyclic graph (DAG) whose vertices are the L labels and whose edges are (π_r, π_j) for every $r < j$. This is a complete DAG relative to the chosen order and therefore contains $L(L-1)/2$ label-to-label edges. An edge in this graph expresses a predictive conditioning relation and should not, in general, be interpreted as a causal relation. In contrast, binary relevance corresponds to the empty label graph, since every label is predicted solely from $\boldsymbol{x}$. These two structures delimit the central trade-off considered in this work: a dense CC makes all preceding labels available to every local classifier, whereas a sparse chain retains only selected conditioning relations. The learning and prediction procedures for the standard dense CC are described next.

2.2. Learning phase
2.3. Prediction phase
2.4. Disadvantages
3. Sparse classifier chains
3.1. Generalities
3.2. Learning phase
3.3. Prediction phase
4. Experiments
4.1. Experimental setting
4.2. Results
5. Conclusion

Figure 1:

Table 1

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CRediT authorship contribution statement

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